

Propiedad de la convolución para exponentes conjugados

Proposición 1. Sean $f \in \mathcal{L}^p(\mathbb{R}^n)$, $g \in \mathcal{L}^q(\mathbb{R}^n)$ con $p, q \in [1, \infty]$ conjugados, entonces

- (1) $f * g \in \mathcal{L}^\infty(\mathbb{R}^n)$ y $\|f * g\|_\infty \leq \|f\|_p \|g\|_q$.
- (2) $f * g$ es uniformemente continua en $\mathbb{R}^n$.

Demostración:

- (1) Aplicando la desigualdad de Hölder a la definición de convolución, tenemos que

$$|f * g(x)| = \left| \int_{\mathbb{R}^n} f(x-y) g(y) dy \right| \leq \int_{\mathbb{R}^n} |f(x-y)| |g(y)| dy \leq \|f\|_p \|g\|_q.$$

Por tanto, $\|f * g\|_\infty \leq \|f\|_p \|g\|_q$.

- (2) Definimos $\tilde{f}(x) := f(-x)$ y consideramos $p \in [1, \infty)$. Por el [lem-convergencia-lp-traslacion/??](#), dado $\varepsilon > 0$, existe $\delta > 0 : \forall h \in B(0, \delta) : \left\| T_h \tilde{f} - \tilde{f} \right\|_p < \frac{\varepsilon}{\|g\|_q}$. Sean $a, b \in \mathbb{R}^n$ tales que $\|a - b\| < \delta$, entonces

$$\begin{aligned} |f * g(b) - f * g(a)| &= \left| \int_{\mathbb{R}^n} (f(b-y) - f(a-y)) g(y) dy \right| \\ &\leq \int_{\mathbb{R}^n} |T_b \tilde{f}(y) - T_a \tilde{f}(y)| |g(y)| dy \\ &\stackrel{\text{Hölder}}{\leq} \left(\int_{\mathbb{R}^n} |T_b \tilde{f}(y) - T_a \tilde{f}(y)|^p dy \right)^{\frac{1}{p}} \|g\|_q \\ &= \left(\int_{\mathbb{R}^n} |T_a (T_{b-a} \tilde{f} - \tilde{f})(y)|^p dy \right)^{\frac{1}{p}} \|g\|_q \leq \frac{\varepsilon}{\|g\|_q} \|g\|_q = \varepsilon. \quad \blacksquare \end{aligned}$$

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